

IAS PRELIMS/MAINS

Ecology & Environment

A stylized illustration of a green city skyline with various skyscrapers, wind turbines, and a sun, set against a light green background with a globe-like shape.

A COMPREHENSIVE COVERAGE

MODULE - 1

BASIC CONCEPTS - 1

ENVIRONMENT

- 1** Environment is the Physical surrounding in which biotic (living) and abiotic (non-living) factors interact with each other.
- 2** All organisms are dependent on the environment to carry out their natural life processes (birth to death) and to meet their physical requirements (food, energy, water, oxygen, shelter etc.).
- 3** The environment is not static. Both biotic and abiotic factors are dynamic and keep changing continuously.



HABITAT

- Habitat is the environment in which an organism lives.
- A single habitat may be common for more than one organism which has similar requirements.
- For example, a single aquatic habitat may support a fish, frog, crab, phytoplankton and many other kinds of organisms.



fish

frog



crab

All habitats are part of the environment but all the places in the environment are not habitats.

DIFFERENCE BETWEEN HABITAT AND ENVIRONMENT?

- A habitat always has life in it, whereas the environment does not necessarily have life in it.
- There are certain places in the environment like the summit of Mt. Everest or the Mariana Trench where the living conditions are extreme and life cannot exist.





CHINA

Learning Space

East China Sea

North Pacific Ocean

MYANMAR (BURMA)

VIETNAM

Lao PDR

THAILAND

CAMBODIA

Gulf of Thailand

South China Sea

Philippine Sea

Mariana Trench

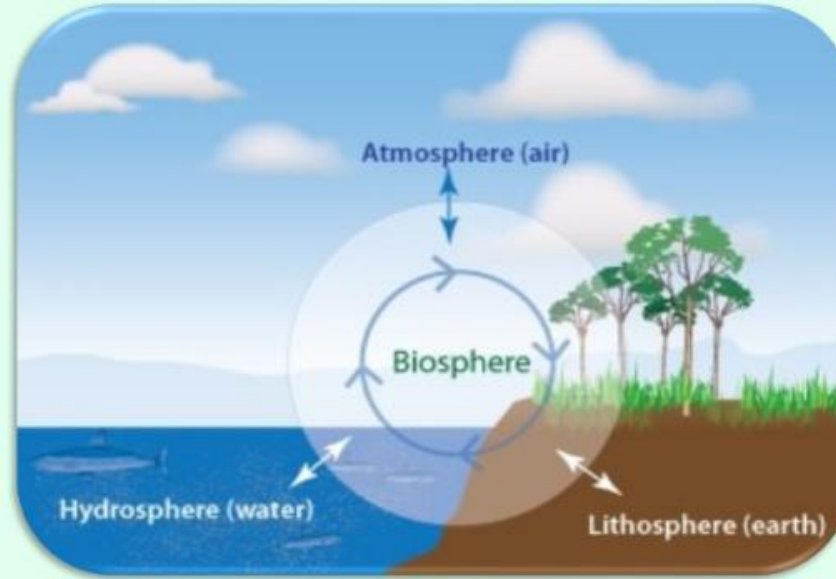
Papua New Guinea

Solomon Sea

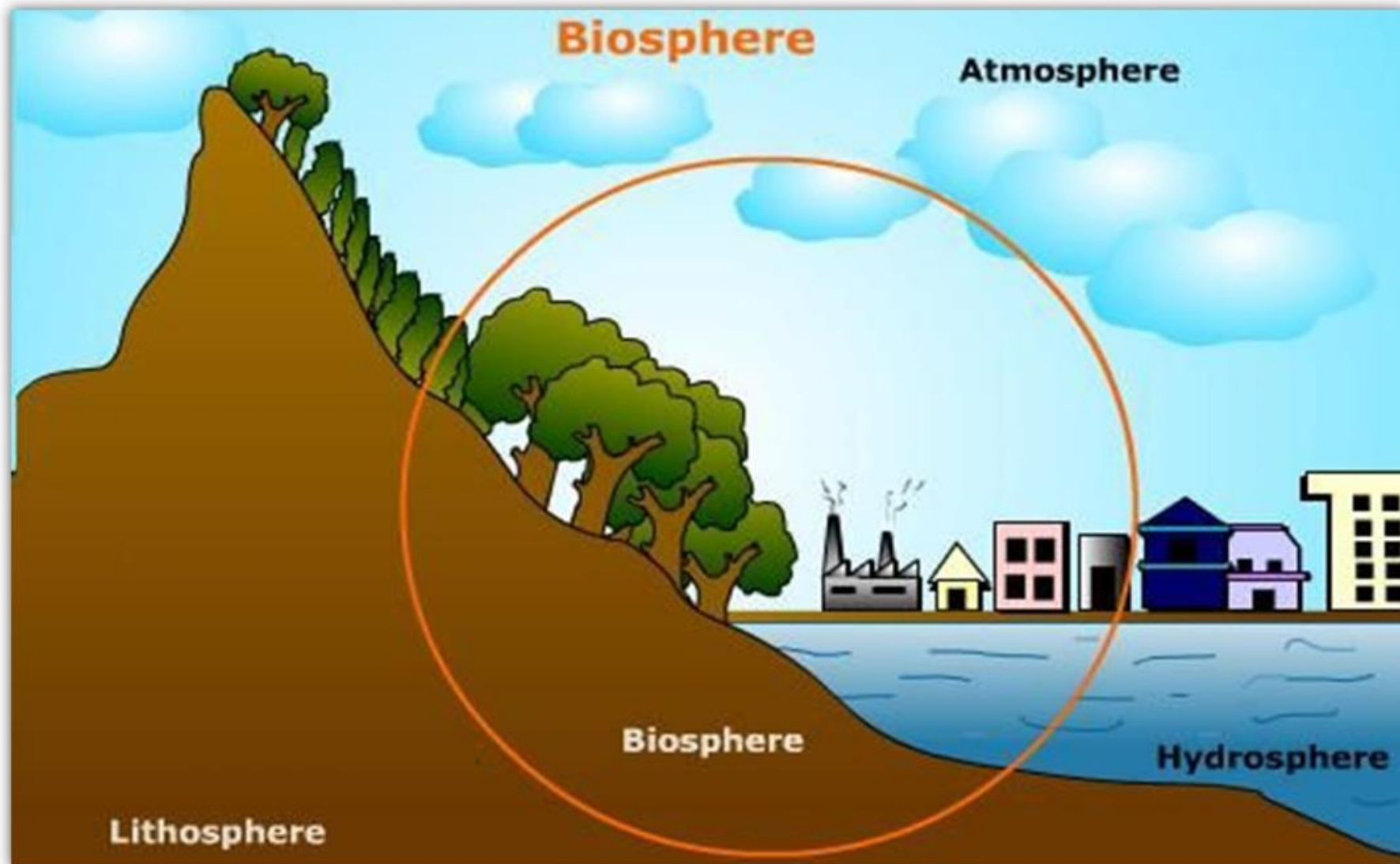
Coral Sea

BIOSPHERE

The biosphere is the biological component of earth which includes all living organisms, their interaction with lithosphere, hydrosphere and atmosphere, together with the dead organic matter produced by them.



BIOSPHERE



ECO SYSTEM

- An ecosystem is the functional unit of nature, where living organisms interact among themselves and also with the surrounding environment.
- Ecosystem varies greatly in size from a small pond to a large forest or a sea.

ECO SYSTEM

TERRESTRIAL ECOSYSTEM

- Forest
- Grassland and
- Desert



AQUATIC ECOSYSTEM

- Pond
- Lake and
- River



MAN - MADE ECOSYSTEM

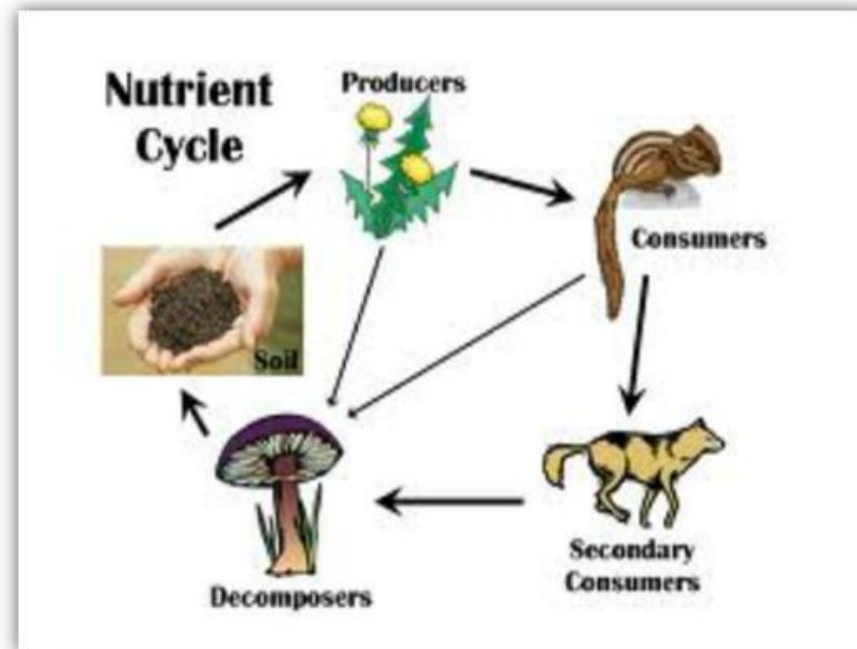
- Crop fields and
- Aquarium



NUTRIENT CYCLES / BIOGEOCHEMICAL CYCLES

In the ecosystem, biotic and abiotic components are linked together through nutrient cycles and energy flows through food chains.

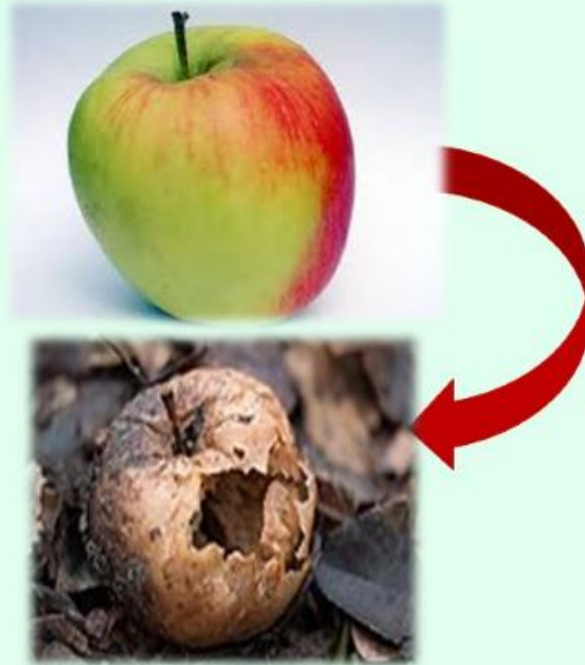
- Biogeochemical cycles are pathways by which nutrients flow between the biotic and abiotic components of the Earth.
- The abiotic portion of the Earth includes
 - ✓ the Lithosphere,
 - ✓ the Hydrosphere and
 - ✓ the Atmosphere.



Many of the nutrients that living things depend on, such as carbon, nitrogen, and phosphorous are in constant circulation through these cycles.

DECOMPOSITION

- Decomposition is a crucial step in the Nutrient Cycles.
- Decomposition is the process by which organic substances are broken down into simpler matter.
- Decomposition is essential for recycling the finite matter in the environment.



WE EXPRESS OUR
SINCERE THANKS
FOR VIEWING THIS VIDEO

Presented by



Learning Space

For Suggestions:

suggestions@learningspace.in

To Contact us:

info@learningspace.in

For Information:

info.learningspacedigital@gmail.com

Visit us at:

www.learningspacedigital.com

0866 244 44 72

98499 42299

94415 27588